Extension of Myopic Coverage Path Planning from Rover to UAV (Technical Summary & Research Proposal Outline)

This document summarizes the engineering modifications required to adapt the algorithm presented in “Risk-Aware Coverage Path Planning for Lunar Micro-Rovers Leveraging Global and Local Environmental Data” to an Unmanned Aerial Vehicle (UAV). The goal is to maintain the lightweight myopic coverage strategy while extending the system to operate in a 3D exploration environment with aerial vehicle dynamics.

1. Key Structural Changes

Component	Original Rover System	Proposed UAV System
State Space	2D grid (x,y)	3D voxel grid (x,y,z)
Neighbor Set	8 neighboring cells	26 neighboring voxels
Terrain Risk	DEM slope cost	Altitude change + obstacle clearance risk
Energy Model	Wheel rolling energy	Thrust■based flight energy model
Obstacle Avoidance	Bug algorithm	Artificial Potential Field (APF)
SLAM	HDL Graph SLAM	LIO■SAM (LiDAR + IMU)
Map Representation	2D occupancy grid	3D voxel map

2. UAV Mathematical Models

Altitude Risk Cost

$$mc\_alt = w1|z(t+1) - z(t)| + w2(1 / d\_obs) + w3|z(t+1) - z\_ref|$$

Energy Consumption Model

$$E\_move = P\_hover * t + m*g*(z(t+1) - z(t)) + F\_drag * d$$

Hover Power Model (Momentum Theory)

$$P\_hover = (m*g)^{(3/2)} / sqrt(2 * rho * A)$$

Final UAV Cost Function

$$mc = \alpha(mc\_static + mc\_visited * V\_i) + \beta\ mc\_alt + \gamma\ mc\_energy$$

3. Algorithm Flow

```
UAV Sensors (LiDAR + IMU)
        ↓
LIO-SAM SLAM
        ↓
3D Voxel Map
        ↓
Myopic Coverage Planner
        ↓
Compute Cost for 26 Neighbor Voxels
(static + visited + altitude risk + energy)
        ↓
Artificial Potential Field Obstacle Avoidance
        ↓
Select Minimum Cost Voxel
        ↓
UAV Motion Controller
```

4. UAV Coverage Algorithm Pseudocode

```
Initialize 3D voxel map
Initialize UAV position

while coverage < target:

    sense nearby voxels
    update occupancy map

    neighbors = generate 26 neighbors

    for each neighbor:
        compute motion cost
        cost = static + visited + altitude + energy

    choose neighbor with minimum cost

    apply APF obstacle avoidance
    move UAV to selected voxel

    update SLAM pose

end
```

5. Expected Research Contributions

• Extension of myopic coverage path planning to 3D UAV exploration environments.
• Introduction of altitude■aware risk cost functions for aerial navigation.
• Integration of energy■aware motion planning based on UAV thrust models.
• Development of a lightweight 3D coverage planner compatible with real■time UAV systems.
• Implementation in simulation environments such as MATLAB UAV Toolbox, Gazebo, or

CoppeliaSim.